

Clustering

Given a set of N points $x^1, x^2, \dots, x^N$
a distance function between points
 $d(x^i, x^j) \rightarrow \mathbb{R}^+$

* Many variants of clustering methods

* we will focus on 2 variants

- ① - Flat grouping of the N points into k clusters
(given k)
- ② - Probabilistic grouping where each group
defines a distribution over x .

Flat clustering:
 Given $D = \{x^1, \dots, x^N\}$, $d(x^i, x^j)$, k

Goal: find $c(x^i) \rightarrow [1 \dots k] \leftarrow$ the group to which i th point belongs.

$m_1, \dots, m_k \leftarrow$ centers or representations of the k -clusters.

such that

one of these objectives is minimized.

$$\textcircled{1} \text{ PD: } \min \sum_{k=1}^k \sum_{\substack{i,j \\ \text{s.t.} \\ c(x_i)=k \\ c(x_j)=k}} d(x^i, x^j)$$

$\leftarrow$ minimizing pairwise distance among points that belong to the same cluster.

$$\textcircled{2} \text{ DC: } \min \sum_{k=1}^k \sum_{i: c(x_i)=k} d(x^i, m_k)$$

$\leftarrow$ distance of points to the cluster center to which point belongs.

show that when $d(x^i, x^j) \equiv$ Euclidean distance
and $m_k = \frac{1}{|C_k|} \sum_{i: c(x^i)=k} x^i$
→ size of cluster k

then objectives (1) & (2) are equivalent

objective ③
 $\max \sum_{\substack{i,j \\ c(x^i) \neq c(x^j)}} d(x^i, x^j) \Leftrightarrow$ objective 1

Algorithm to solve for:

$$\min_{m_1, m_2, \dots, m_k} \min_{\{c(x^i)\}} \left[\sum_{k=1}^k \sum_{i: c(x^i)=k} d(x^i, m_k) \right] \leftarrow \underline{F(m, c)}$$



NP-hard.



Settle for heuristics

—

k-means a well-known greedy algorithm

k-mean algorithm:

Initially: $m_1^0, m_2^0, \dots, m_k^0 \leftarrow$ random points from D .

for $t=1 \dots \infty$

c-update [update the assignments of points to cluster c

for $i=1$ to N

$c^t(x^i) \leftarrow \underset{k=1 \dots k}{\operatorname{argmax}} d(x^i, m_k)$

$d(x^i, m_k)$

$\leftarrow F(m^t, c^t) \leq F(m^{t-1}, c^{t-1})$

μ -update / $\star$ update the centroids of
 for $k=1$ to k
 $m_k^t \leftarrow \operatorname{argmin}_m \sum_{i: c(x^i)=k} d(x^i, m)$
 $F(m^t, c^t) \leq F(m^{t+1}, c^t)$
 for sq. Euclidean distance, the optimal value
 will be $m_k^t = \frac{\sum_{i: c(x^i)=k} x^i}{\sum_{i: c(x^i)=k} 1}$

If $F(m^t, c^t) = F(m^{t+1}, c^{t+1})$
 stop.

the k -means algorithm will terminate in a finite # of steps to a local minima.

Main intuition: with each iteration, each of
 C- μ -update decrease the objective $F(m^t, c^t)$ or

or learn it unchanged.
we stop when $F(m^t, c^t) = F(m^{t-1}, c^{t-1})$

Demo: <https://colab.research.google.com/drive/11OM1gf1X0DNGp0juWfaU0FrADh5hHiP9?usp=sharing>

Probabilistic Clustering

Reading: Chapter: 8.7.2 The EM algorithm,
Chapter 21.4 (Clustering using Mixture models)

Given $D = \{x^1, \dots, x^N\}$

of clusters K .

$f(x|\theta)$ functional form of the density function at each cluster.

Goal:

Find parameter $\theta_1 \dots \theta_K$ of the parameters of each cluster

$\pi_1 \dots \pi_K$ & weights of cluster $\sum_{k=1}^K \pi_k \geq 1$ $\pi_k \geq 0$

such that

the likelihood of the data (as defined below)
is maximized.

$$L(D) = \prod_{i=1}^N P(x^i) = \frac{N}{\pi_1} \underbrace{\sum_{k=1}^K \pi_k f(x^i | \theta_k)}_{P(x^i)}$$

$P(x^i)$ is the mixture
of K distributions.

Example density functions:

$$① f(x|\theta_k) = N(x; \underbrace{\mu_k, \Sigma_k}_{\theta_k})$$

$$② f(x|\theta_k) = \prod_{j=1}^d p(x_j^i | \theta_{kj}) \leftarrow \text{simple independent density}$$

when $k=1$, the objective can be solved in closed form for many simple distributions

eg: Gaussian:

$$\pi^{MLE}, \theta^{MLE} \leftarrow \underset{\pi, \theta}{\operatorname{argmax}} \sum_{i=1}^N \log \sum_{k=1}^K \pi_k f(x^i | \theta_k)$$

$$\text{s.t.} \quad \pi_k \geq 0 \quad \sum_{k=1}^K \pi_k = 1$$

For $k=1$

$$\theta^{MLE} \equiv \mu, \Sigma$$

$$\leftarrow \sum_{i=1}^N x^i / N, \quad \frac{1}{N} \left[\sum_{i=1}^N (x^i - \mu)(x^i - \mu)^T \right]$$

For $k > 1$, closed form solution such as the above is not possible.

Gradient descent is an option in such cases.

But can we do better?

The EM algorithm that we discuss next allows use of such closed form updates.

The main trick that the algorithm exploits is a particular property of log-concave functions provided by the theorem below:

Thm:

$$\log \sum_{z=1}^k g(x^i, z) = \max_{q_{iz}} \left[\sum_{z=1}^k q_{iz} \log g(x^i, z) - \sum_z q_{iz} \log q_{iz} \right]$$

$$\begin{aligned} & \sum_{z=1}^k q_{iz} \log g(x^i, z) - \sum_z q_{iz} \log q_{iz} \\ & \text{s.t. } q_{iz} \geq 0 ; \sum_{i=1}^N q_{iz} = 1 \end{aligned}$$

Proof:

$Q(q, g)$ is concave in q .

solve for the optimal "q" by equating the gradient to "0" of the Lagrangian.

$$Q(q, g, \lambda) \Big|_{q_{iz}} \max \sum_{z=1}^k q_{iz} \log g(x^i, z) - \sum_z q_{iz} \log q_{iz} + \lambda \left(\sum_z q_{iz} - 1 \right)$$

$$\frac{\partial Q}{\partial q_{iz}} = \log g(x^i, z) - \log q_{iz} - \frac{1}{q_{iz}} + \lambda = 0$$
$$\Rightarrow \underline{q_{iz}} \propto g(x^i, z)$$

$$q_{iz}^* = \frac{g(i, z)}{\sum_{k=1}^K g(i, k)}$$

By plugging in the optimal solution in $Q(q, g)$ we can show that LHS = RHS.

Use the above variational trick to rewrite the original objectives as:

$$\max_{\pi_1, \dots, \pi_K} \sum_{i=1}^N \max_{\{q_{iz}\}} \sum_{z=1}^K q_{iz} \log \pi_z f(x^i | \theta_z) - \sum_{z=1}^K q_{iz} \log q_{iz}$$

$$\theta_1, \dots, \theta_K$$

Solving the above objective iteratively.

Initially: $\pi_1^0, \dots, \pi_k^0, \theta_1^0, \dots, \theta_k^0 \leftarrow$ Random parameter values.

for $t = 1$ to ∞ - - {

 A solve for the auxiliary variables z /

 for $i = 1$ to N

$$q_{iz}^t \leftarrow \frac{\pi_z^{t+1} f(x^i | \theta_z^{t+1})}{\sum_{k=1}^K \pi_k^{t+1} f(x^i | \theta_k^{t+1})} g(x^i, z)$$

E-step

$\uparrow$
 probability that x^i belongs to cluster z
 at iteration " t ".

1. Solve for $\pi_1, \dots, \pi_k$ & $\theta_1, \dots, \theta_k$ variables given fixed q_{ik}^t values of auxiliary variables &

for $k = 1$ to k ξ

$$\theta_k^t = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^N q_{ik}^t \log f(x^i | \theta) \leftarrow$$

Example if $f(x^i | \theta) \sim \mathcal{N}(x^i; \mu, \Sigma)$

$$\mu_k^t = \frac{\sum_{i=1}^N q_{ik}^t x^i}{\sum_{i=1}^N q_{ik}^t}$$

$$\Sigma_k^t = \frac{\sum_{i=1}^N q_{ik}^t (x^i - \mu_k^t)(x^i - \mu_k^t)^T}{\sum_{i=1}^N q_{ik}^t}$$

M-step.

$$\pi_k^t = \frac{\sum_{i=1}^N q_{ik}^t}{N}$$

Concluding

- The EM algorithm is guaranteed to converge to the local minima of the original objective.
- Demo: [here.](#)